

01

AI-POWERED EDUCATIONAL LEARNING PLATFORM



DOMAIN

- Artificial Intelligence(AI)
- Education Technology (EdTech)
- Generative AI



PROJECT TYPE

- Full-Stack Web Application
- AI-Based Learning Assistant



OBJECTIVE

- Provide intelligent, document - based learning through Retrieval-Augmented Generation (RAG) and Gemini AI



KEY TECHNOLOGIES

- React+Vite
- Flask REST API
- Gemini AI
- ChromaDB
- Sentence Transformers

02

TEAM DETAILS



01

Member 1

Name: Pracheta Gajbhiye

Role: Project Lead & System Architect

Contribution: Project Planning, architecture design



06

Member 6

Name: Samiksha More

Role: Technical Associate

Contribution: Assisted in coding and implementation



02

Member 2

Name: Srushti Dangre

Role: Backend Developer

Contribution: Flask APIs, authentication, document processing



07

Member 7

Name: Ashwini Lamdade

Role: Project contributor

Contribution: Project contributor



03

Member 3

Name: Kunjal Divecha

Role: Frontend Developer

Contribution: React UI, chat interface ,file upload



08

Member 8

Name: Trisha Dash

Role: Project contributor

Contribution: Project contributor



04

Member 4

Name: Mayuri Sonawane

Role: AI/ML Engineer

Contribution: RAG pipeline,Gemini AI, embeddings, prompt engineering



09

Member 9

Name: Sachi Patil

Role: Project contributor

Contribution: Project contributor



05

Member 5

Name: Pari Shukla

Role: Project contributor

Contribution: Project contributor



10

Member 10

Name: Shravani Gajbhiye

Role: Project contributor

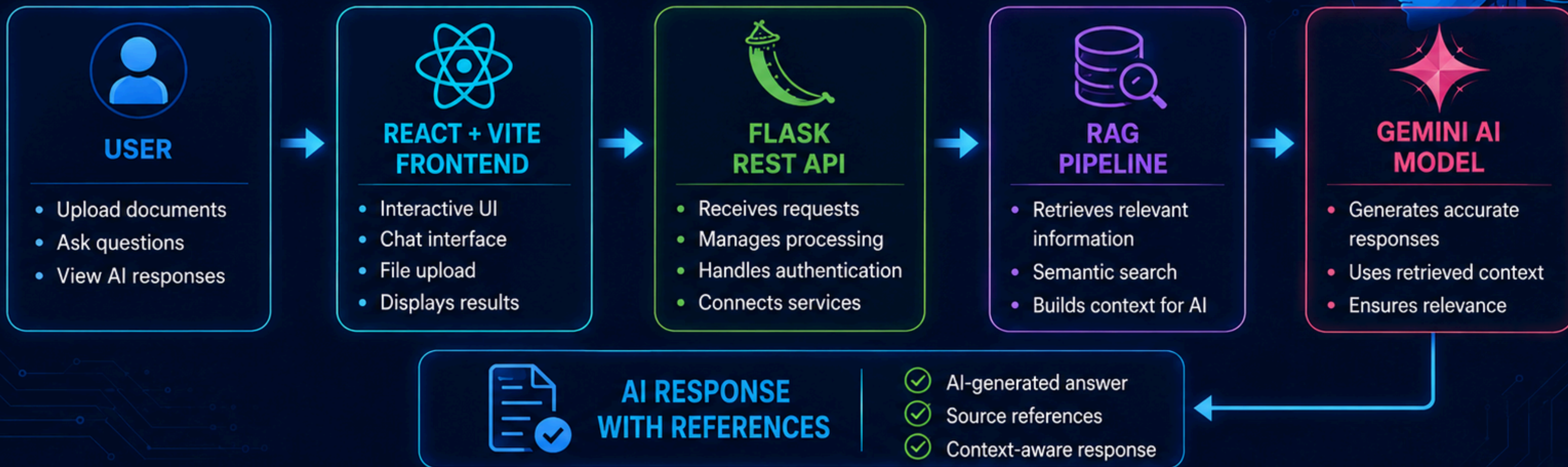
Contribution: Project contributor

03

PROJECT ARCHITECTURE

(HIGH-LEVEL)

SYSTEM WORKFLOW



TECHNOLOGIES USED



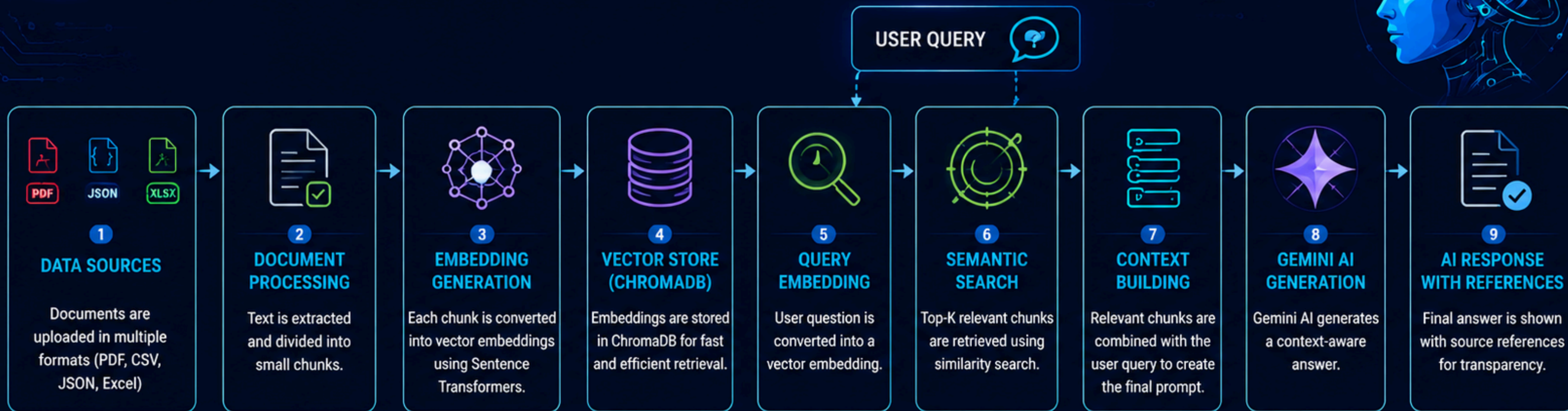
KEY BENEFITS



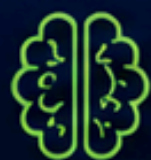
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RAG PIPELINE ARCHITECTURE

HOW OUR AI UNDERSTANDS & ANSWERS YOUR QUESTIONS



TECHNOLOGIES POWERING THE PIPELINE



Sentence Transformers
For text embeddings



ChromaDB
Vector Database for semantic search



Gemini AI
Model for intelligent answer generation.



Flask
REST API for backend



React + Vite
Frontend for interactive UI

KEY BENEFITS



Accurate & context-aware answers



Source-backed responses for transparency



Works with multiple data formats



Scalable, fast & reliable

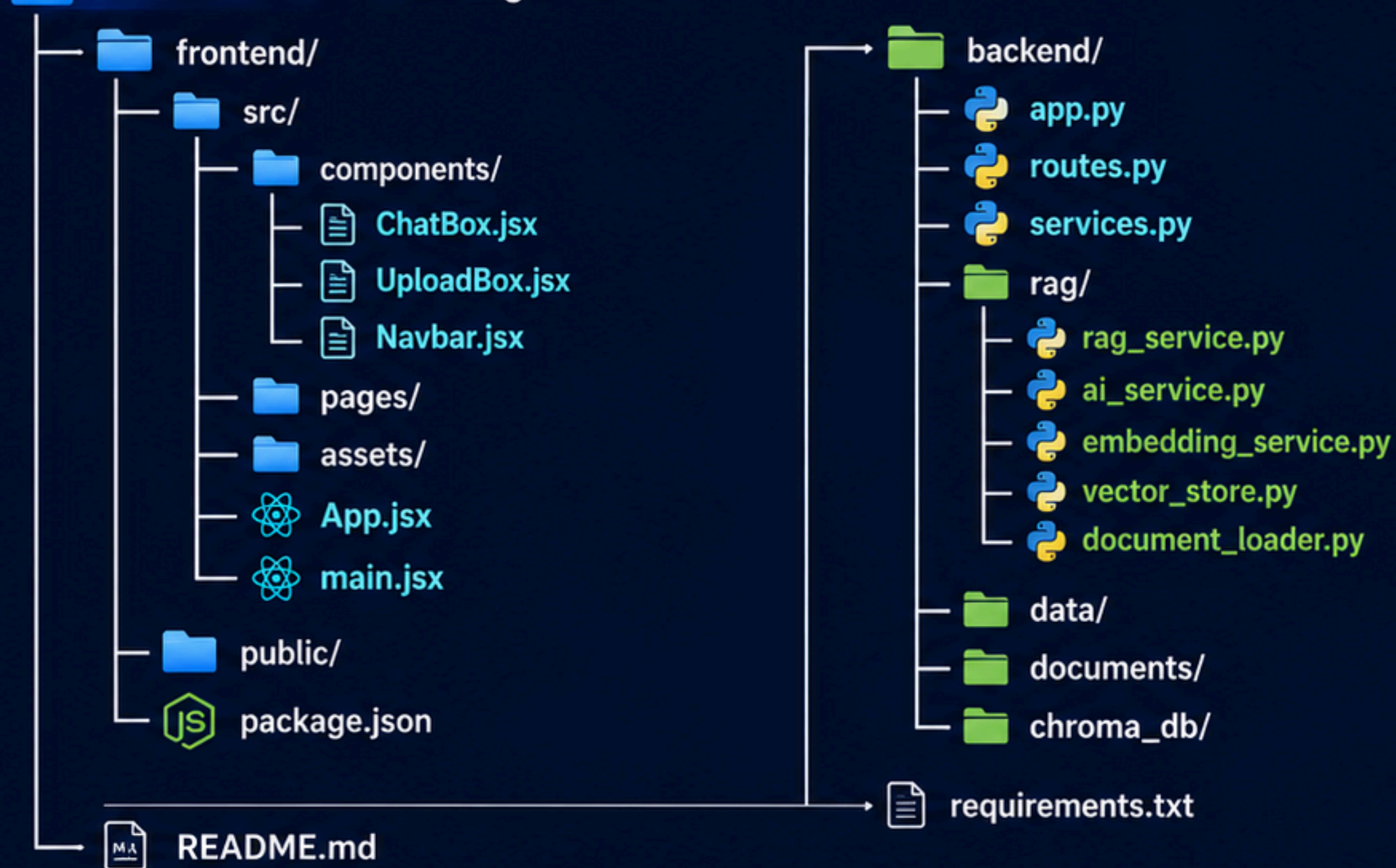
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GITHUB DIRECTORY STRUCTURE

Organized • Scalable • Easy to Maintain



AI-Educational-Learning-Platform/



KEY FILES & PURPOSE

 App.jsx	Main React Application
 ChatBox.jsx	AI Chat Interface
 UploadBox.jsx	Document Upload Component
 app.py	Flask Entry Point
 routes.py	API Endpoints
 services.py	Backend Services
 rag_service.py	RAG Pipeline Orchestration
 ai_service.py	Gemini AI Integration
 embedding_service.py	Generate Embeddings
 vector_store.py	ChromaDB Operations
 document_loader.py	Load & Process Documents



FRONTEND

- React + Vite
- Modern UI/UX



BACKEND

- Flask REST API
- Python Services



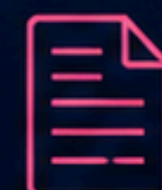
AI & RAG

- Gemini AI
- RAG Pipeline



VECTOR DB

- ChromaDB
- Semantic Search



DATA SOURCES

- PDF
- CSV
- JSON
- Excel